Gravity Books Data Ware House Documentation

- Conceptual Data Modelling

There is an issue in the gravity books store analysis team that they can't do their work smoothly because of the lack of source of data to do the analysis on Therefore, the proposed solution was creating a data mart for the sales modules so they can keep track of the sales, and orders delivery process and do the needed reports in time.
The reporters that the business asks for are to get the total order price and total fee (order price + shipping cost) per month, per shipping method, per author and per country.
And also the business wants to keep track of average days of process and days to deliver per shipping method. To achieve this business needs

- Logical Data Modelling

To achieve what businesses want the most suitable Architecture for this project is **Kimball architecture** Due to the need for only one data mart for the sales module with two fact tables to get the different types of analysis.
The structure of the data mart will be a **Galaxy Schema with two fact tables:**

- Transactional fact: to track sales
- Accumulating snapshot fact: to track delivery process

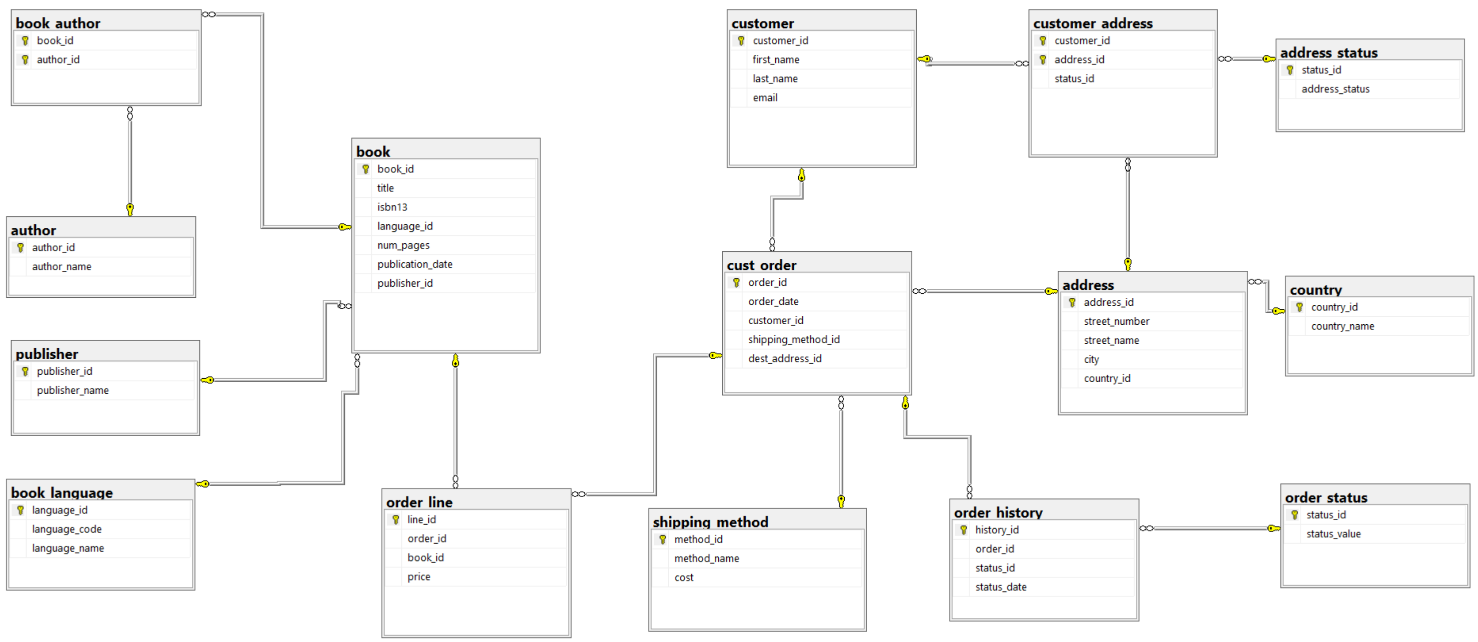
The types of dimensions used will be:

- Conformed Dimension for Date_Dim
- Degenerate Dimension for OrderID, LineID and HistoryID
- Junk Dimension for Status_Dim
- Snow flaked Dimension for Customer to Address and Book to Author as (many to many)
- Role Playing Dimension for Date_Dim with the Accumulating snapshot fact table
- Slowly Changing Dimension Type2 for Shipping_Dim to track changes in shipping cost

After integrating the data into the data mart we build a cube for much faster analysis.

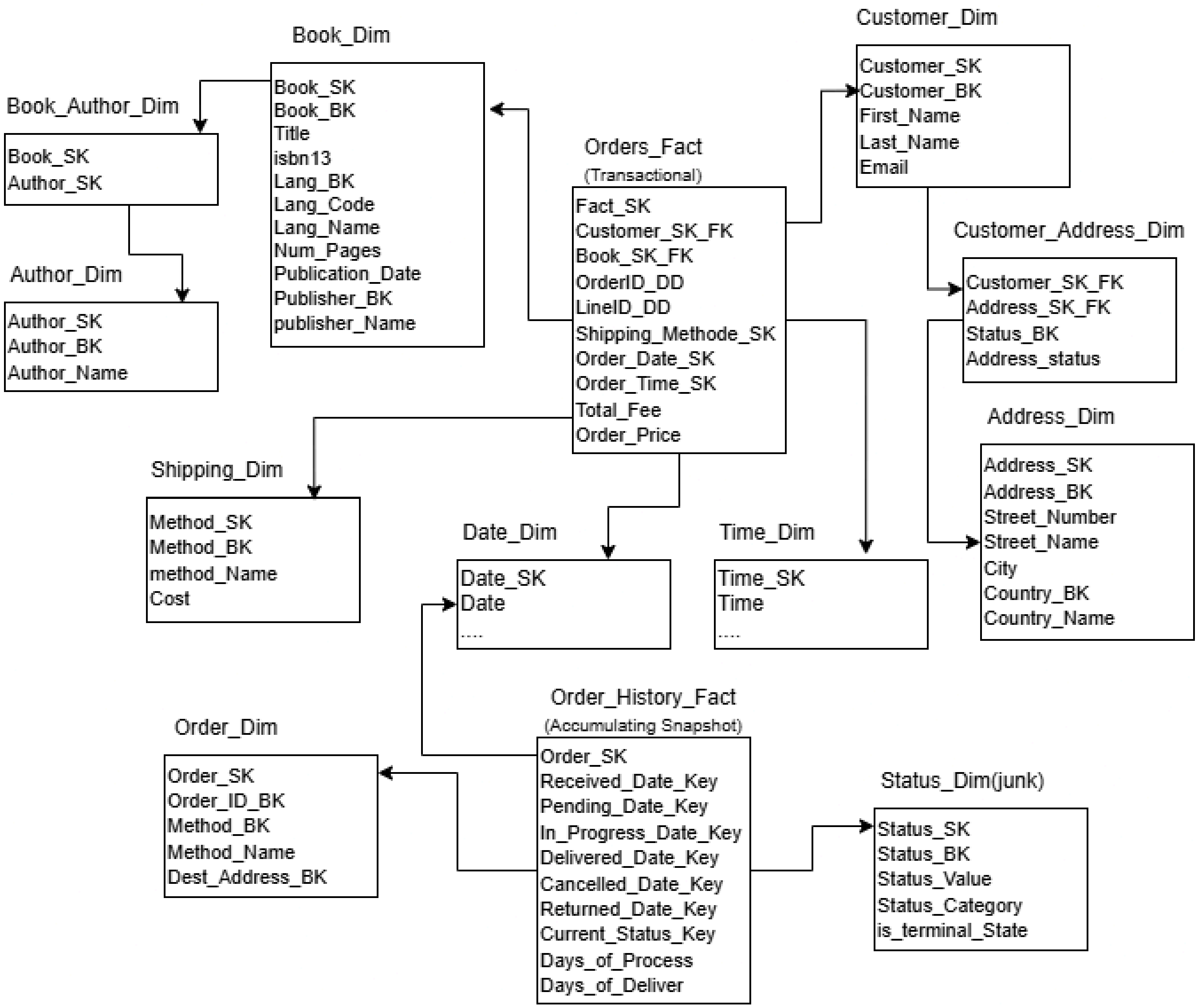
first cube for sales with total fee and total order price as measures and the need dimensions based on business requirement are month, quarter shipping method, author and country.
And the second cube with average days of process and average days to deliver as measures and shipping method dimension.

This is the OLTP Diagram:



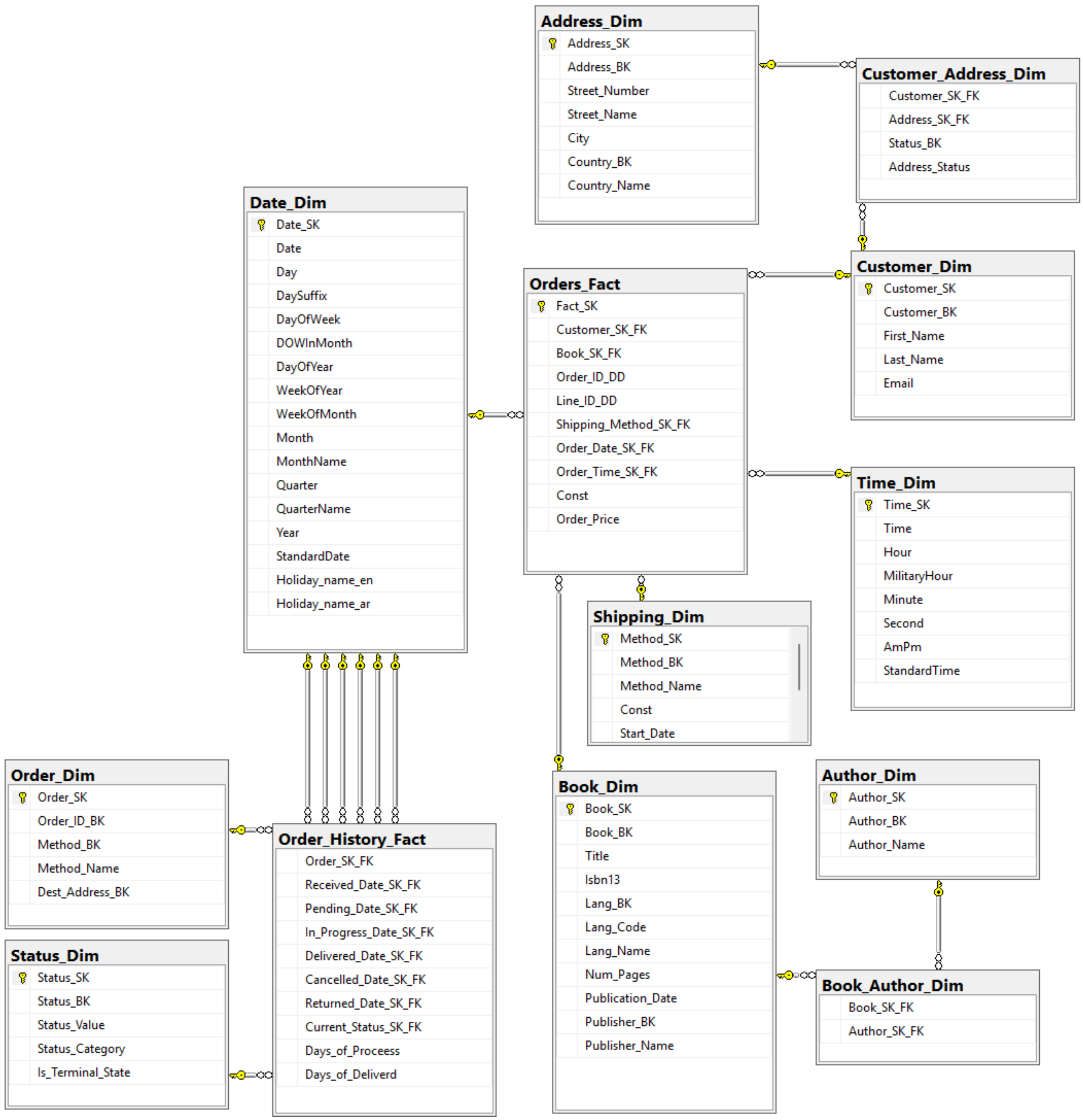
OLAP Solution:

GBs DWH Data Governance Sheet



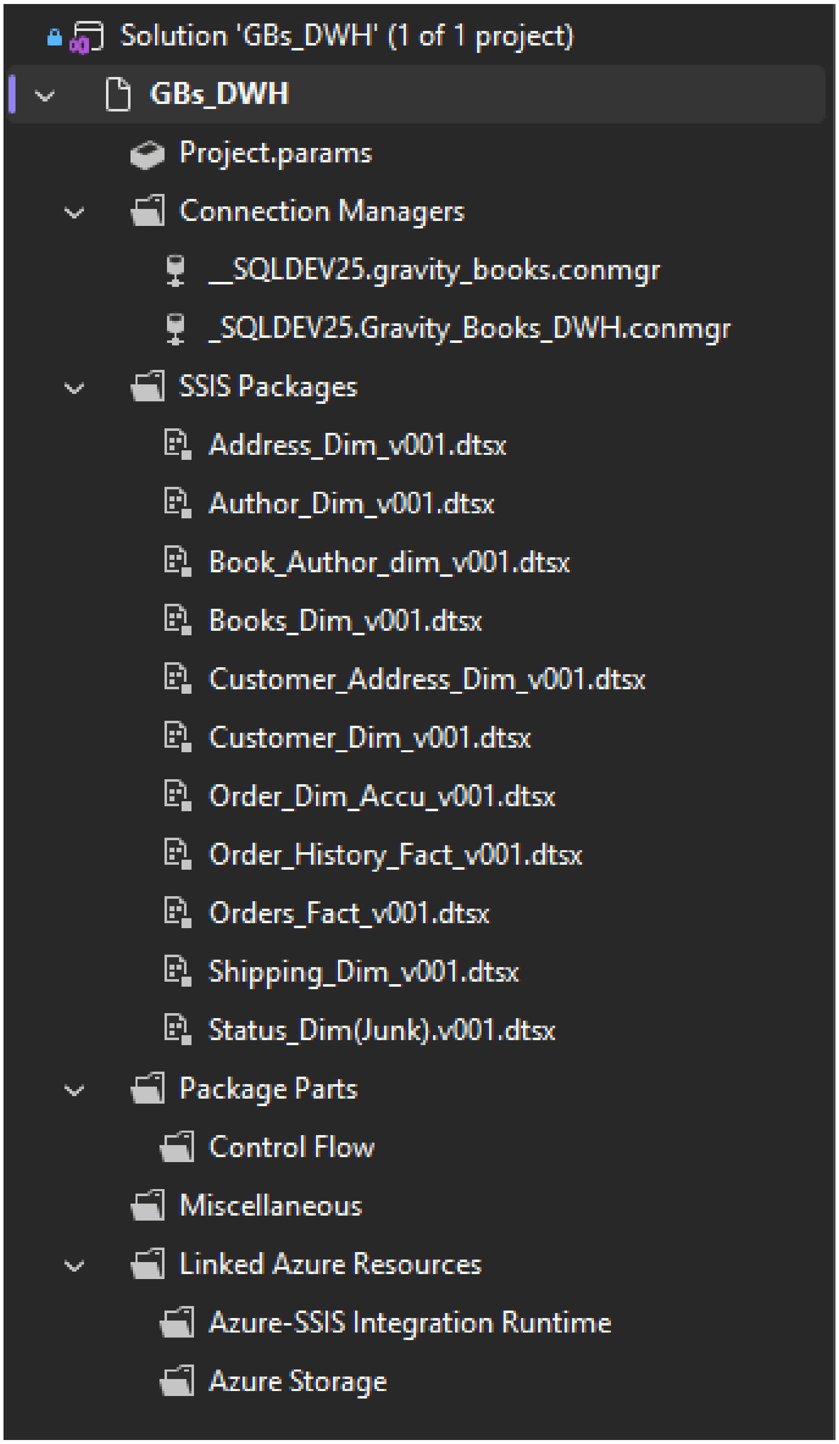
- Physical Data Modelling

The physical implementation of the solution:

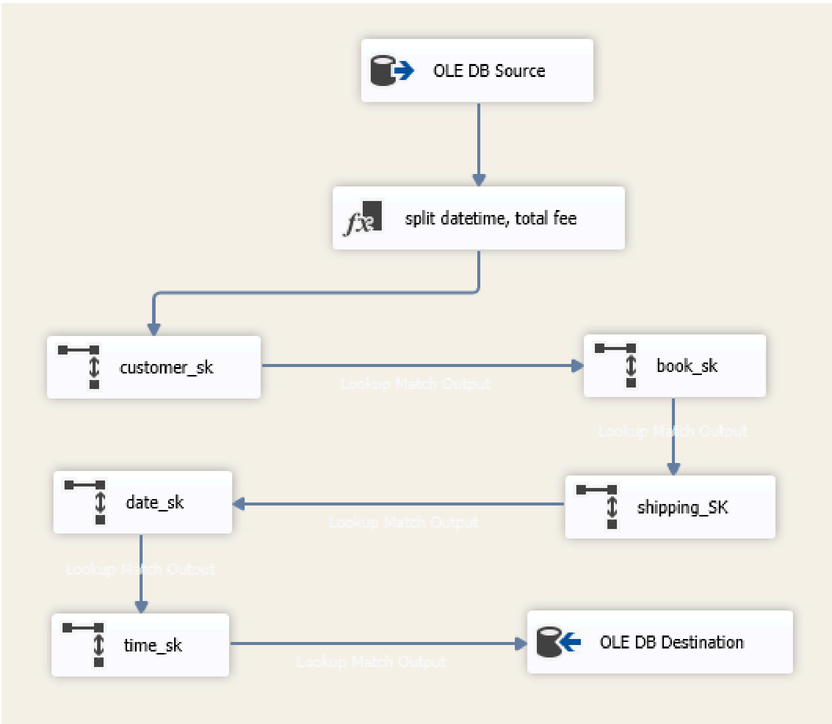


Then the data integrated from OLTP to OLAP using SSIS:

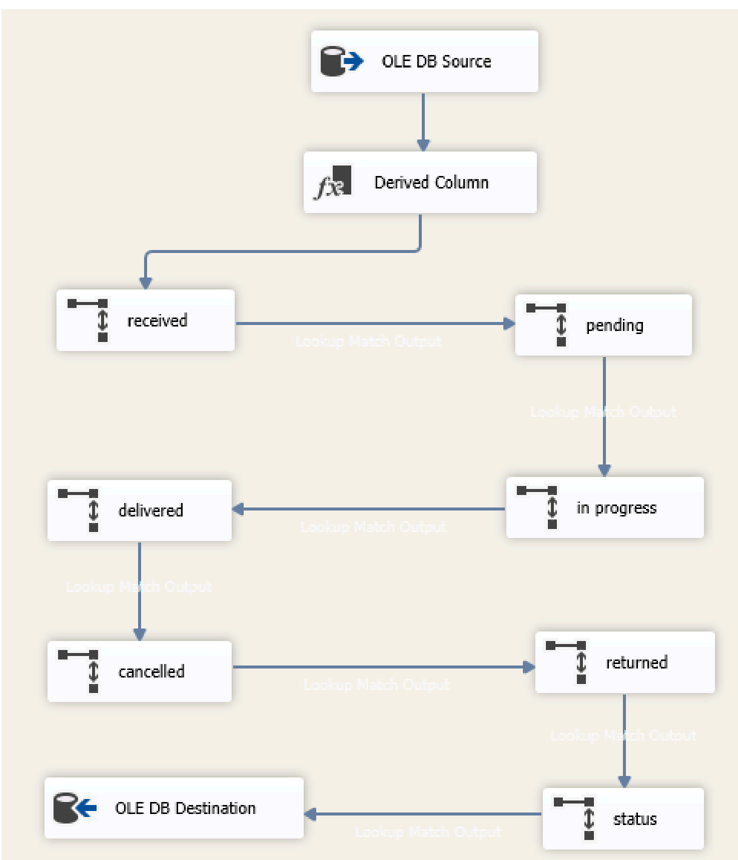
Creating package for each table integration:



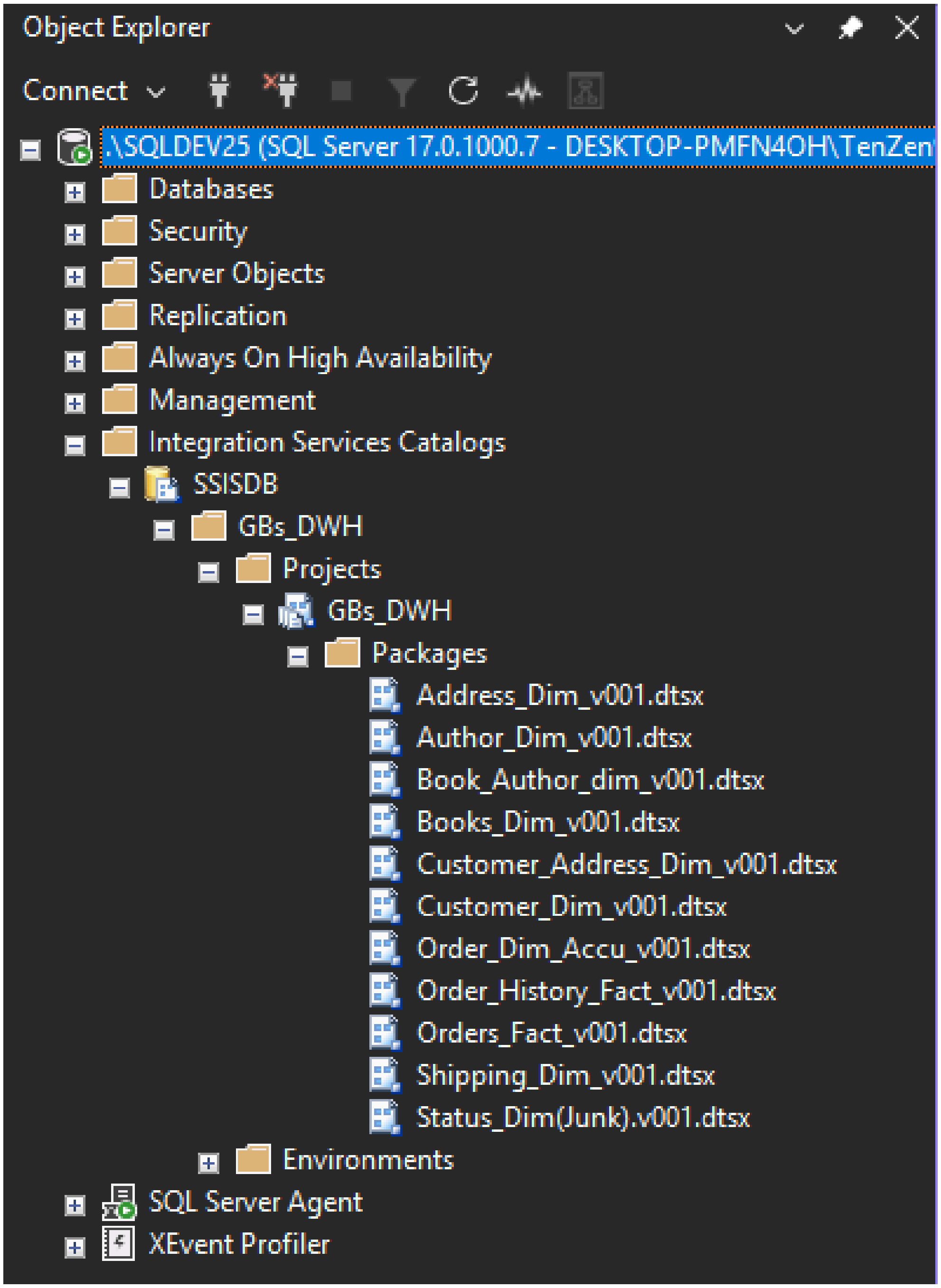
Order Fact Integration:



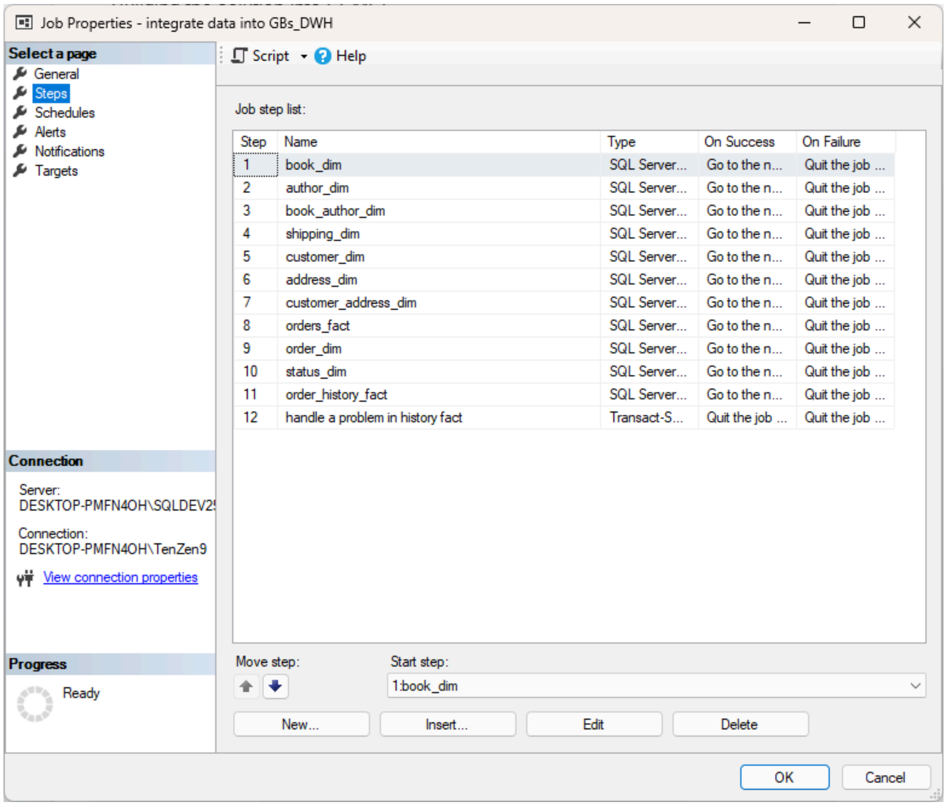
Order Fact History Integration:

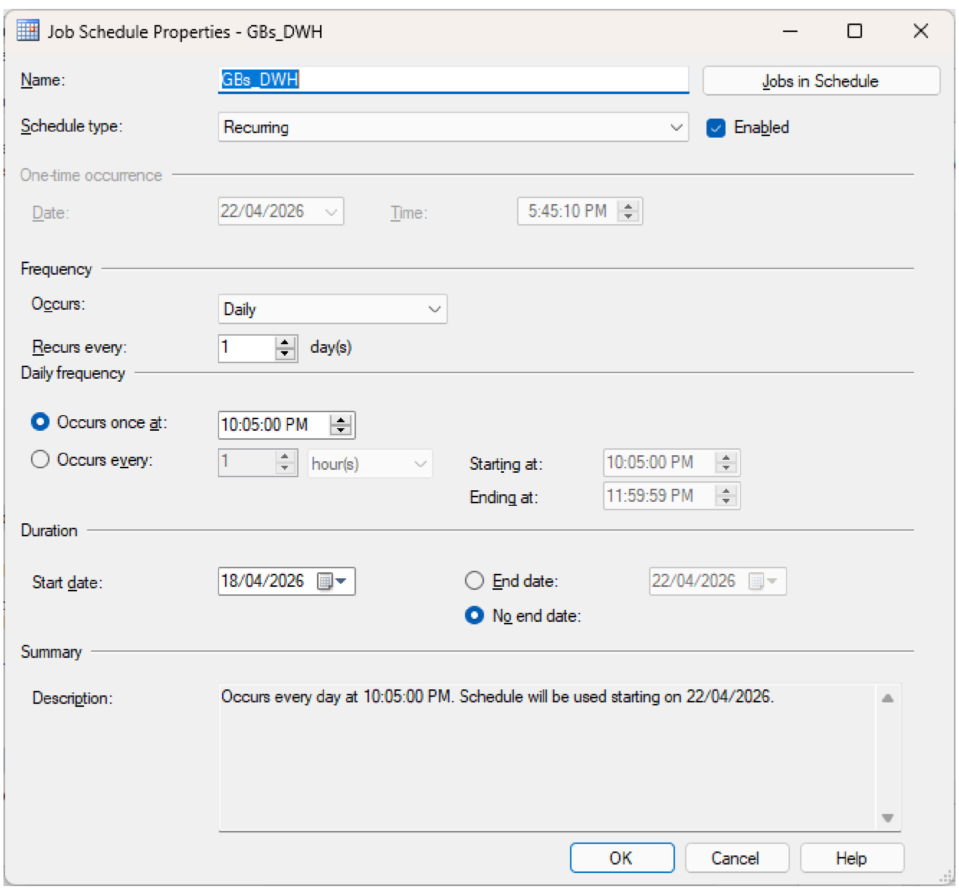


Building the solution into SSMS:

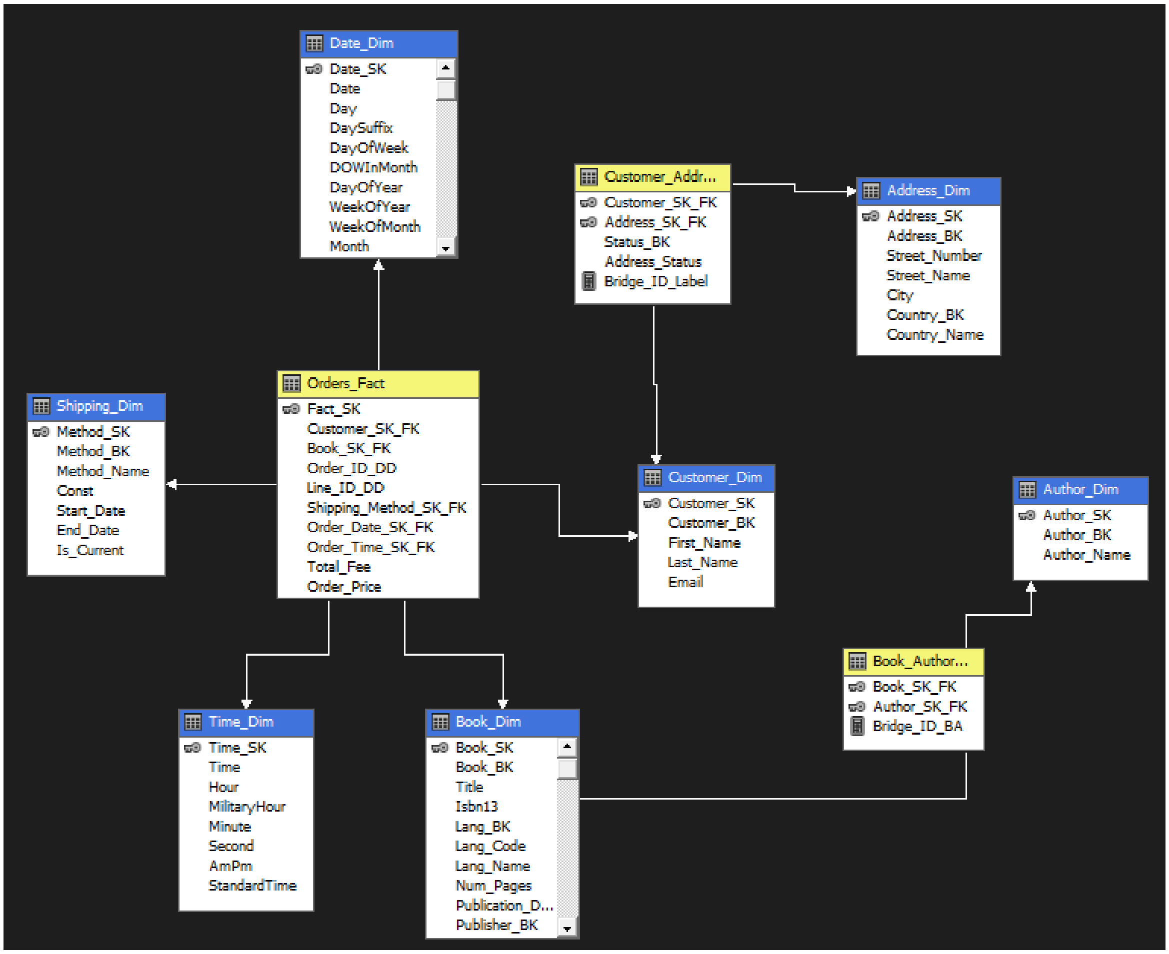


The steps of the scheduling:





Building the cube with SSAS:



Creating the needed analysis by the business with Power BI: